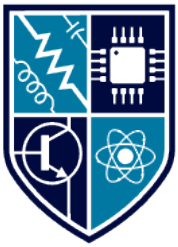
Team Charter

WSGC 2027 Collegiate Rocket Launch Competition - Madison Aeronautics and Space Team (MAST)

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MADISON COLLEGE

Electronics & Electrical Engineering Technology

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Major Team Goals

- The team shall design, build, and fly a high-powered (L1) rocket.
- The team shall learn and apply the engineering design process and principles
- The team shall develop and demonstrate skills for working as a team through an engineering design and prototyping cycle.
- The team shall develop and demonstrate engineering skills through a hands-on approach.
- The team shall develop and demonstrate project management, technical communication, and collaborative problem-solving skills.
- The team shall establish an environment that encourages learning, constructive feedback, and continuous improvement.
- The team shall…*additional goals identified through survey responses*….

Major Deliverables

- The team shall produce a single-stage boosted-dart rocket with a powered 2-4" outside diameter booster structure and an unpowered, passive release separable dart that shall not exceed the 4,000-foot Above Ground Level (AGL) ceiling while maximizing the separation distance between booster and dart.
- The team shall integrate a comprehensive avionics and video recording system to determine and record on-board flight data throughout the flight.
- The team shall incorporate a recovery system for both the booster and dart sections as well as the engine of the rocket.
- The team shall select an approved rocket motor from the list provided by WSGC appropriate to accomplish the competition and engineering goals of the rocket system.
- The team shall document system requirements, design decisions, verification methods with results, and simulation methods with results in the appropriate project repositories.
- The team shall deliver a project presentation as a team that is enthusiastic and informative.
- The team shall deliver the appropriate reports (PDR, CDR, FRR, PLAR) completely, on time, and to maximize the scoring opportunity.
- The team shall propose and deliver an appropriate educational outreach activity/opportunity.

Defining Criteria for Assessing Team and Individual Success

Individual Success – Metrics for Each Individual Member of the Team

- Individuals shall ***support the individual success of other team members***.
- Individuals shall maintain a positive and constructive attitude.
- Individuals shall treat all team members with respect and patience.
- Individuals shall actively participate and collaborate in relevant focus areas.
- Individuals shall consistently and clearly communicate thoughts and ideas with all collaborating team members.
- Individuals shall actively listen and acknowledge contributions of all team members.
- Individuals shall strive to produce quality work that meets or exceeds requirements.
- Individuals shall develop knowledge and skillsets that align with project requirements and solutions.
- Individuals shall record their efforts, progress, and relevant technical findings in the appropriate project repositories.
- Individuals shall strive to complete delegated tasks on time.
- Individuals shall manage time effectively and communicate availability constraints that may affect their responsibilities.
- Individuals shall communicate blockers, anticipated delays, and risks to project progress as early as reasonably possible.
- Individuals shall seek assistance and clarification when necessary.
- Individuals shall accept constructive criticism and incorporate relevant feedback into their work.

Team Success – Metrics for all Collective Members of the Team

- The team shall ***seek unanimity in decisions made*** by building consensus.
- The team shall maintain and actively utilize recognized communication channels and shared project repositories.
- The team shall manage the project effectively and efficiently.
- The team shall solve technical and non-technical problems collaboratively.
- The team shall produce and review all deliverables collaboratively.
- The team shall delegate tasks strategically, accounting for individual strengths, interests, and availability.
- The team shall be respectful in its critique of individual ideas and proposed solutions.
- The team shall evaluate proposed solutions against defined requirements, project constraints, and available technical evidence.
- The team shall account for individual strengths and interests when assigning responsibilities.
- The team shall advance its collective technical skillsets and knowledge.
- The team shall produce quality outputs that meet or exceed requirements.
- The team shall learn from each other and share relevant knowledge and experience.
- The team shall maintain a shared understanding of project status, priorities, dependencies, and outstanding work.
- The team shall identify, communicate, and address project risks, blockers, and unresolved issues before they jeopardize project deliverables.

Costs

- The team shall primarily use funds awarded to Madison College by WSGC.
- The team shall submit required project budget proposals to the program advisor for approval before purchases are needed, so Madison College procedures can be followed.
- The team shall consider budget constraints and expected project benefits when proposing expenditures.
- The team shall obtain all required approvals ***before*** committing funds or making personal purchases.
- The team shall compile receipts for project-related expenses in a shared data repository for review by the program advisor.
- Purchases made by team members for project-related purposes without approval of the program advisor are not guaranteed to be reimbursed.
- Purchases made with Madison College funds shall follow applicable institutional purchasing procedures and tax-exemption requirements.
- Unanimous team approval of a budget proposal shall not replace the required approval of the program advisor or Madison College.

Communications – Applicable to ALL Team Members

- Recognized channels of communication are ***Microsoft Teams***, ***Outlook***, and ***GitHub***.
- The General channel of the WSGC Rocket Competition team in Microsoft Teams shall be the main channel for team broadcasts (messages sent to all team members).
- Broadcasts requiring team member response shall ***be responded to within 1 business day***.
- Project-related communication from any team member on recognized channels shall ***require an initial response within 1 business day***.
- Team members shall attend scheduled project-related meetings and events unless an absence has been scheduled or otherwise communicated.
- Team members shall ***communicate scheduled absences at least 1 business day ahead*** of project-related meetings and events whenever practicable.
- Absent team members shall be personally responsible for closing information gaps following a missed meeting or event, including reviewing meeting notes, decisions, and assigned action items.
- Team members shall communicate anticipated delays, availability constraints, and other circumstances that may affect their assigned responsibilities.
- Team members shall be respectfully critical of ideas and solutions presented by fellow team members.
- Respectful criticisms of ideas and solutions shall be constructive and made in good faith.
- Respectful criticisms of ideas and solutions shall not be given or taken personally.
- Team members shall communicate urgent safety concerns and time-critical project issues promptly through an appropriate communication channel.
- Team members shall document project decisions, technical findings, and action items in the appropriate shared repository when required for project coordination or traceability.

Define Process for Decision-Making and Addressing Conflicts

- The team shall ***seek unanimous agreement*** in all decisions whenever reasonably achievable.
- The team shall consider the *competition requirements*, *due date timeline*, and *budget constraints* when making *all decisions.*
- The team shall also consider technical feasibility, system interfaces, verification requirements, and applicable safety requirements when evaluating proposed solutions.
- The definition and outline of high-level system requirements shall have the ***unanimous approval*** of all team members.
- The team shall document approved system requirements in a shared repository and maintain them as the reference for design, implementation, and verification.
- Proposed changes to approved system requirements shall be reviewed by the team for impacts on project scope, schedule, budget, safety, and affected focus areas before approval.
- The design and verification definitions of solution elements in each focus area shall have the input and ***unanimous approval*** of all team members *within* the respective focus group.
- The design and verification definitions of solution elements that affect multiple focus areas shall have the input and ***unanimous approval*** of all team members *within* each affected focus group.
- Cross-focus-area decisions shall account for system interfaces, dependencies, and the effects of the proposed solution on the overall system.
- An individual tasked with implementing a properly defined and approved design element shall have freedom to develop the implementation as they see fit, so long as design requirements remain achievable, the scope of work fits within the associated timeline, budgetary constraints are not exceeded, and outcomes are verifiable through team review.
- An individual implementing an approved design element shall not independently introduce changes that affect approved requirements, system interfaces, safety, budget, or schedule without the appropriate team review and approval.
- Designed subsystems whose solution elements meet defined requirements and pass verification in each associated focus area shall be reviewed by all team members before integration. Each team member shall have the opportunity to raise objections and suggest revisions, and ***all objections shall be acknowledged and considered before integration proceeds***.
- Verification results and supporting documentation shall be made available for review before a subsystem is approved for integration.
- Any changes made to a subsystem after verification or integration approval shall be reviewed to determine whether additional verification or approval is required.
- All required project budget proposals shall receive ***unanimous approval*** by the team and be sent to the program advisor for approval.
- ***When unanimity cannot be reached***, the team shall identify the unresolved concerns, review the applicable requirements and constraints, and attempt to reach agreement through further discussion. It is appropriate to disagree but to give consensus in order for the team to move on.
- ***Unresolved disagreements*** that affect project requirements, task assignments, timelines, budget, system interfaces, or integration shall be escalated to the entire team and, when necessary, to the program advisor and associated faculty advisor for resolution.
- No team decision shall override applicable competition rules, safety requirements, institutional policies, or the authority of the program advisor, faculty advisors, or launch-site safety personnel within their respective responsibilities.
- Any serious conflict between team members in one or more focus areas, including project

management, shall be promptly acknowledged and escalated by another team member, preferably one in a project management role, directly to the program advisor or appropriate faculty advisor. **Escalation to the program advisor is required in any serious conflict scenario.**

- Team members shall attempt to resolve ordinary disagreements respectfully and collaboratively, but serious conflicts shall not be left unresolved while awaiting team consensus.
- Decisions resulting from escalated conflicts shall be communicated to affected team members and documented when they affect project requirements, responsibilities, schedule, budget, or other project commitments.

Project Management and Accountability

- The project management team shall coordinate project planning, task assignments (*"Is your task properly defined?"*), schedule tracking (*"Do you have enough time?"*), communication (*"Do you need help?"*), and progress reporting (*"Is there complete confidence when you say you are done?"*).
- The project management team shall maintain awareness of project dependencies, risks, and blockers that may affect successful completion of deliverables.
- The team shall collaboratively define task responsibilities, expected outcomes, and due dates clearly enough for progress and completion to be assessed.
- Team members shall maintain accurate task status and communicate changes that may affect agreed-upon commitments.
- The project management team shall facilitate collaboration and accountability but shall not independently override approved technical decisions or requirements.
- The program and faculty advisors shall provide guidance (*"Do you need help?"*) and resolve disputes regarding requirements (*"Is there complete confidence when you say you are done?"*), timelines (*"Do you have enough time?"*), task assignments (*"Is your task properly defined?"*), and other matters that exceed the project management team's ability to resolve.
- The team shall retain collective responsibility for its engineering work and decisions within the authority granted by the competition, WSGC, Madison College, and the program and faculty advisors.

Charter Review and Commitment

- The team shall review this charter collaboratively before adopting it as the team's working agreement.
- All team members shall understand the expectations and processes defined in this charter and shall be responsible for following them. They shall indicate such by their initials.
- The team shall review this charter when significant changes to team organization, project scope, or competition requirements warrant revision.
- Proposed changes to this charter shall be discussed and receive acknowledgement from all team members and advisors before adoption.
- This charter shall remain subject to applicable competition requirements, safety requirements, program advisor determination, and Madison College policies.

Version History

Version	Date	Author	Change-log
1.0.0	10/20/2026	F.A.Filip	Initial document
1.0.1	10/20/2026	G.R.Emmel	Initial review/edits